

Package ‘EMGCR’

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Type Package

Title Fit a Mixture Cure Rate Model With Custom Link Function

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Description Tools to fit Mixture Cure Rate (MCR) models via the Expectation-Maximization (EM) algorithm, allowing for flexible link functions in the cure component and various survival distributions in the latency part. The package supports user-specified link functions, includes methods for parameter estimation and model diagnostics, and provides residual analysis tailored for cure models.

License GPL-3

Encoding UTF-8

Depends R (>= 3.5),
survival,
Formula,
dplyr,
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actuar

Imports survival,
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tibble

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liver

Liver cancer data

Description

A sample of 2,766 patients who have been diagnosed with liver cancer between 2012 and 2016 whose cancer grades are well identified. Available individual-level covariates include age at diagnosis, sex, grade of liver cancer, count of relapse, and median household income. The grade of the disease is categorized into four levels: well-differentiated (Grade I), moderately differentiated (Grade II), poorly differentiated (Grade III), and undifferentiated/anaplastic (Grade IV).

Usage

```
data(liver)
```

Format

This data frame contains the following columns:

- time: survival time in month
- status: censored = 0, dead = 1
- sex: 1 if male, 0 if female
- age: Age at diagnosis
- meth: Median household income of the subject
- relapse: Count of relapses after first diagnosis
- grade: Histologic grade of liver cancer

Examples

```
data(liver)
head(liver)
```

MCRfit

Fit a Mixture Rate Cure (MRC) Model with Custom Link Function

Description

This function fits a survival model to the provided data using a specified link function, with options for weights, iteration limits, and tolerance.

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Usage

```

MCRfit(
  formula,
  data,
  dist = "weibull",
  link = "logit",
  tau = 1,
  maxit = 1000,
  tol = 1e-05
)

MCRfit(
  formula,
  data,
  dist = "weibull",
  link = "logit",
  tau = 1,
  maxit = 1000,
  tol = 1e-05
)

```

Arguments

<code>dist</code>	A character string. The distribution function to use. Defaults to "weibull". Options include "Exponential", "Rayleigh", "Log-normal", "Log-logistic" or "invgauss".
<code>link</code>	A character string. The link function to use. Defaults to "logit". Options include "logit", "probit", "plogit", "rplogit" or "cauchit".
<code>tau</code>	A numeric value. A threshold or cutoff parameter. Defaults to 1.
<code>maxit</code>	An integer. The maximum number of iterations for the optimization procedure. Defaults to 1000.
<code>tol</code>	A numeric value. The convergence tolerance for the optimization procedure. Defaults to 1e-5.
<code>time</code>	A numeric vector. The observed survival times.
<code>status</code>	A numeric vector. The censoring indicator, where 1 typically indicates an event (e.g., death) and 0 indicates right censoring.
<code>x</code>	A numeric matrix or data frame. Covariates for the survival model.
<code>w</code>	A numeric vector. Weights for each observation. Defaults to equal weights if not provided.

Details

This function implements a custom survival model fitting procedure, allowing the user to specify a link function and other model parameters. It supports weighted observations and iterative optimization with user-defined tolerance and iteration limits.

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Value

A list containing:

coefficients	The estimated coefficients for the covariates.
convergence	A logical value indicating whether the optimization converged.
iterations	The number of iterations performed.
logLik	The log-likelihood of the fitted model.

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coefficients	The estimated coefficients for the covariates.
convergence	A logical value indicating whether the optimization converged.
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logLik	The log-likelihood of the fitted model.

Examples

```
data(liver)
#head(liver)
model1 <- MCRfit(Surv(time,status)~age+medh+relapse+grade|-1+sex+age+medh+grade,data=liver)
summary(model1)
data(liver)
#head(liver)
model1 <- MCRfit(Surv(time,status)~age+medh+relapse+grade|-1+sex+age+medh+grade,data=liver)
summary(model1)
```

qqMCR

QQ-Plot of Residuals for MCR Model

Description

QQ-Plot of Residuals for MCR Model

Usage

```
qqMCR(
  object,
  type = c("cox-snell", "quantile"),
  envelope = FALSE,
  nsim = 100,
  censor = NULL,
  ...
)
```

Arguments

object	An object of class "MCR"
...	Additional arguments (not used)

residuals.MCR	<i>Compute residuals for MCR model</i>
---------------	--

Description

This function computes Global Cox-Snell and randomized quantile residuals for objects of class MCR.

Usage

```
## S3 method for class 'MCR'
residuals(object, type = c("cox-snell", "quantile"), ...)
```

Arguments

object	An object of class MCR, typically returned from MCRfit .
type	Type of residual.
...	Additional arguments (not used).

Value

A numeric vector of residuals.

Examples

```
data(liver)
#head(liver)
model <- MCRfit(Surv(time,status)~age+medh+relapse+grade|-1+sex+age+medh+grade,data=liver)
summary(residuals(model,type="quantile"))
```

rMCM	<i>Generate Random Samples for Mixture Cure Rate (MCR) Model</i>
------	--

Description

Simulates survival data from a mixture cure rate model with covariates and user-defined link and latency distributions. Censoring is applied randomly.

Usage

```
rMCM(
  n,
  x,
  w,
  censor,
  alpha,
  beta,
  eta,
  dist = "weibull",
```

```

    link = "logit",
    tau = 1
  )

```

Arguments

n	Integer. Number of observations to simulate.
x	Matrix or numeric. Covariate matrix for the latency component (must include intercept if needed).
w	Matrix or numeric. Covariate matrix for the cure component (no intercept assumed).
censor	Numeric. Maximum censoring time (uniformly distributed).
alpha	Numeric. Shape parameter for the survival distribution.
beta	Numeric vector. Coefficients for the latency part.
eta	Numeric vector. Coefficients for the cure part.
dist	Character. Distribution for the latency part. Options: "weibull", "lognormal", "loglogistic", "invgauss", "exponential", "rayleigh".
link	Character. Link function for cure component. Options: "logit", "probit", "rplogit", "cauchit".
tau	Numeric. Tuning parameter for power link functions. Defaults to 1.

Value

A list with elements:

time Observed (possibly censored) survival time.

status Event indicator (1 = event, 0 = censored).

x Covariate matrix for the latency component.

w Covariate matrix for the cure component.

pCcensur Percentage of cured individuals.

pUCcensur Percentage of censored cases among the uncured.

Examples

```

# Example: Simulating survival data using the inverse Gaussian distribution
library(EMGCR)

n <- 500
beta <- c(1, -1, -2)
eta <- c(0.5, -0.5)
alpha <- 1.5

p <- length(beta)
q <- length(eta)

set.seed(10)
X <- matrix(rnorm(n*(p-1),0,1),n,p-1)
X <- cbind(1,X)

set.seed(20)
W <- matrix(runif(n*q,-1,1),n,q)

```

```
W <- scale(W)

max_censoring <- 10

set.seed(1234)
sim_data <- rMCM(n=n, x = X, w = W,
                 censor = max_censoring,
                 beta = beta, eta = eta,
                 alpha = alpha,
                 link = "logit", dist = "invgauss", tau = 1)

names(sim_data)
head(sim_data)
attributes(sim_data)
attr(sim_data, "pCcensur")
attr(sim_data, "pUCcensur")
```

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